

Steps

After reviewing the resources for this assignment, do the following steps...

1: Start a Word document, making sure you start with your name, the assignment name and the due date.

2: On your PC or on the Windows VirtualBox installation, open the Command Prompt.

- As an alternative, with the Windows Key pressed, tap R for the Run Window, and type in cmd
- Type Exit to leave the Command Prompt
- In the Search Area, type cmd, and the Command Prompt should be an option
- Take a screenshot of the Screen and insert it into the document with the caption Command Prompt
- Open PowerShell using either method above, except using PowerShell in place of cmd
- Take a Screenshot of the screen and insert it into the document with the caption PowerShell screen
- Perform each of the Windows commands listed in the table below. Each time take a screenshot and insert it into the document with a caption of which command was used and what it did.
- When finished, type exit to close the terminal Window.

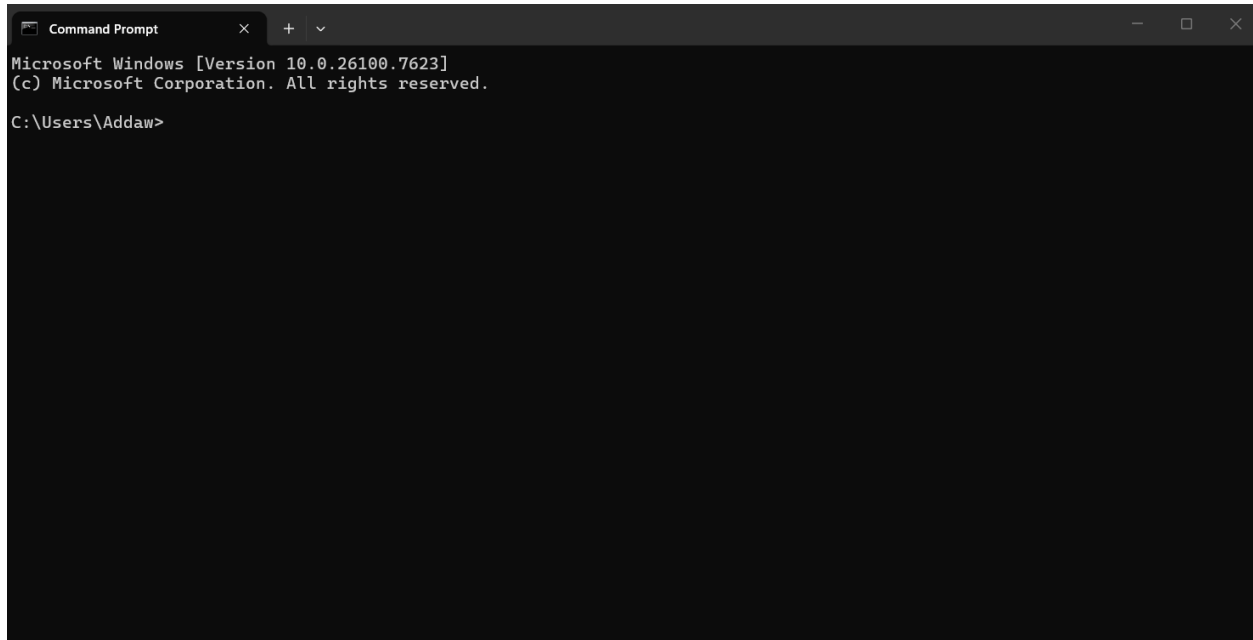
3: In the Linux system on your VirtualBox, open the terminal. How to open a terminal window varies by operating system. Take a screenshot of the Linux Terminal and

- Perform each of the Linux commands listed in the table below. Each time take a screenshot and insert it into the document with a caption of which command was used and what it did.
- When finished, close the terminal window.

Note: For Mac users, the terminal window in your OS will operate just like a Linux Terminal window since MacOS is a close relative of Linux based on the Unix Kernel just like Linux is.

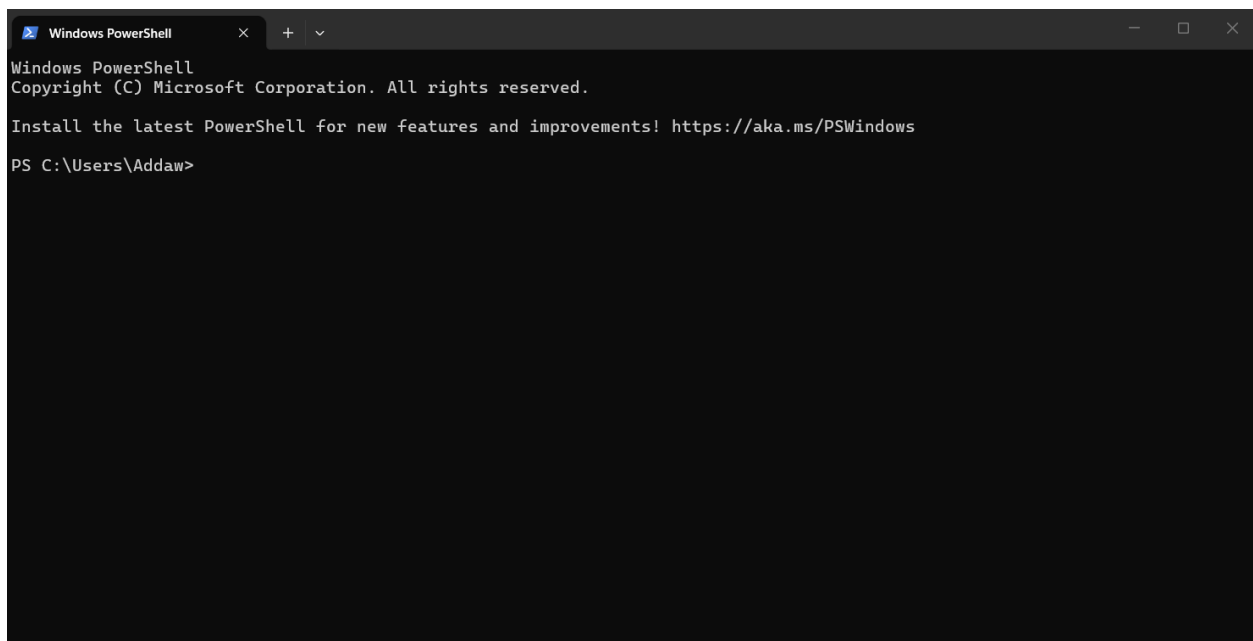
Using my own PC system:

Command Prompt:

A screenshot of a Windows Command Prompt window. The title bar shows 'Command Prompt' with standard window controls. The text inside the window reads: 'Microsoft Windows [Version 10.0.26100.7623] (c) Microsoft Corporation. All rights reserved. C:\Users\Addaw>'. The prompt is at the end of the line, ready for input.

```
Microsoft Windows [Version 10.0.26100.7623]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Addaw>
```

PowerShell screen:

A screenshot of a Windows PowerShell window. The title bar shows 'Windows PowerShell' with standard window controls. The text inside the window reads: 'Windows PowerShell Copyright (C) Microsoft Corporation. All rights reserved. Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows PS C:\Users\Addaw>'. The prompt is at the end of the line, ready for input.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Addaw>
```

Dir: Shows the contents of the directory you are currently in.



Command Prompt



C:\Users\Addaw>dir

Volume in drive C is Windows

Volume Serial Number is B29C-B2F3

Directory of C:\Users\Addaw

File Name	Size	Attributes	Creation Date	Creation Time
.		<DIR>	02/10/2026	07:20 AM
..		<DIR>	03/17/2025	01:43 AM
.anaconda		<DIR>	10/27/2025	03:27 PM
.bash_history	36		09/25/2024	02:17 PM
.cache		<DIR>	12/12/2025	06:28 PM
.conda		<DIR>	12/13/2025	06:13 AM
.condarc	25		10/27/2025	03:18 PM
.continuum		<DIR>	10/27/2025	03:17 PM
.eclipse		<DIR>	10/04/2024	01:20 PM
.gitconfig	254		09/18/2024	02:06 PM
.ipynb_checkpoints		<DIR>	12/12/2025	11:00 PM
.ipython		<DIR>	11/10/2025	03:27 PM
.jupyter		<DIR>	11/21/2025	10:44 AM
.kaggle		<DIR>	11/21/2025	11:17 AM
.matplotlib		<DIR>	12/12/2025	06:28 PM
.p2		<DIR>	04/20/2025	07:00 PM
.packettracer	176		05/07/2025	09:14 PM
.VirtualBox		<DIR>	02/11/2026	11:39 AM
.vscode		<DIR>	09/25/2024	02:09 PM
anaconda3		<DIR>	10/27/2025	03:19 PM
anaconda_projects		<DIR>	11/03/2025	01:47 PM
BrawlhallaReplays		<DIR>	03/22/2024	09:30 PM
Cisco Packet Tracer 8.2.2		<DIR>	05/08/2025	06:48 PM
Contacts		<DIR>	03/17/2025	01:49 AM
Documents		<DIR>	04/24/2025	08:49 PM
Downloads		<DIR>	02/09/2026	10:40 PM
eclipse		<DIR>	09/10/2024	03:10 PM
eclipse-workspace		<DIR>	04/20/2025	07:01 PM
Favorites		<DIR>	03/17/2025	01:49 AM
FirstProject		<DIR>	02/22/2024	04:33 PM
github		<DIR>	09/11/2024	01:52 PM
hazalldr22.github.io		<DIR>	09/18/2024	01:55 PM
Links		<DIR>	03/17/2025	01:49 AM
Music		<DIR>	12/16/2025	12:32 PM
obs-studio		<DIR>	04/21/2025	05:26 PM
OneDrive		<DIR>	02/10/2026	10:58 AM
Saved Games		<DIR>	03/17/2025	01:49 AM
Searches		<DIR>	03/17/2025	01:49 AM
Untitled.ipynb	72		10/27/2025	03:27 PM
Untitled1.ipynb	72		10/27/2025	03:27 PM
Untitled2.ipynb	72		11/21/2025	10:44 AM
Untitled3.ipynb	628		12/12/2025	11:02 PM
Videos		<DIR>	04/21/2025	09:28 PM
VirtualBox VMs		<DIR>	02/09/2026	10:47 PM

8 File(s) 1,335 bytes
36 Dir(s) 271,893,045,248 bytes free

Cat: The cat command doesn't work on the Windows command prompt. Fortunately, the "type" command is the cat alternative for cmd. In this case I output a log from VirtualBox (It ended up being super long as you can see from the scroll bar on the left side). You may need to zoom in to see everything.

```
Command Prompt
The directory name is invalid.

C:\Users\Addaw\VirtualBox VMs\Windows 10 (OS)\logtype VBox.log
00:00:19.180888 VirtualBox VM 7.2.4 r170995 win.amd64 (Oct 17 2025 12:31:09) release log
00:00:19.180815 Log opened 2025-02-11T16:36:43.136000000Z
00:00:19.180817 Build type: release
00:00:19.180821 OS Product: Windows 11
00:00:19.181212 OS Release: 19.0.26108.7623
00:00:19.181224 OS Service Pack:
00:00:19.671293 VM Product Name: HP Pavilion x360 Convertible 15-eh0xx
00:00:19.586795 VM Product Version: Type\ProductConfigId
00:00:19.586763 Firmware type: UEFI
00:00:19.587389 Secure Boot: Enabled
00:00:19.587525 Host RAM: 16351MB (15.7GB) total, 6048MB (6.0GB) available
00:00:19.587811 Executable: C:\Program Files\Oracle\VirtualBox\VirtualBox.exe
00:00:19.587833 Process ID: 25812
00:00:19.587834 Package type: 62B0D5C_64BIT_GENERIC
00:00:19.587835 Windows features:
00:00:19.587836 Core Isolation (Memory integrity): DISABLED
00:00:19.590964 Installed Extension Packs:
00:00:19.598128 Oracle VM VirtualBox Extension Pack (Version: 7.2.4 r170995; VMDK Module: VBoxVMDK; Crypto Module: VBoxPaaS(Crypto))
00:00:19.592811 Console Machine state changed to 'Restoring'
00:00:19.592811 GUI: Qt version: 6.8.0
00:00:19.592878 GUI: H2D 120v type is enabled
00:00:19.613720 GUI: UPMediumEnumerator: Medium-enumeration finished!
00:00:19.778190 SUP: sup ss: R 0x00000000 18 0x00010000
00:00:19.778197 SUP: sup s1: R 0x00010000 18 0x00100000
00:00:19.778197 SUP: sup s2: R 0x00100000 18 0x00050000
00:00:19.778197 SUP: sup s3: R 0x00050000 18 0x00010000
00:00:19.778197 SUP: sup s4: R 0x00010000 18 0x00010000
00:00:19.778197 SUP: sup s5: R 0x00010000 18 0x00010000
00:00:19.778197 SUP: sup s6: R 0x00010000 18 0x00010000
00:00:19.778197 SUP: sup s7: R 0x00010000 18 0x00010000
00:00:19.778197 SUP: sup s8: R 0x00010000 18 0x00010000
00:00:19.787151 SUP: Loaded VMX86 r0 (C:\Program Files\Oracle\VirtualBox\VMX86.r0) at 0000000000000000 - ModuleInit at 0000000000000000 and ModuleTerm at 0000000000000000 using the native ring-0 loader
00:00:19.787176 SUP: winhlpd reload // C:\Program Files\Oracle\VirtualBox\VMX86.r0=0x0000000000000000
00:00:19.787194 SUP: winhlpd reload // C:\Program Files\Oracle\VirtualBox\VMX86.r0=0x0000000000000000
00:00:19.793238 Guest architecture: amd
00:00:19.793239 Guest OS type: 'Windows10_64'
00:00:19.796760 VMX86CPUID: No raw-mode support in this build!
00:00:19.796761 Using execution engine 1
00:00:19.807111 File system of 'C:\Users\Addaw\VirtualBox VMs\Windows 10 (OS)\snapshots' (snapshots) is stfs
00:00:19.807111 File system of 'C:\Users\Addaw\VirtualBox VMs\Windows 10 (OS)\Windows 10 (OS).vdi' is stfs
00:00:19.810806 File system of 'C:\Users\Addaw\Downloads\Windows.iso' (DVD) is stfs
00:00:19.880877 Shared Clipboard: Initialized OK
00:00:19.880879 Shared Clipboard: Service loaded
00:00:19.880879 Shared Clipboard: Mode: Off
00:00:19.880880 Shared Clipboard: Service running in normal mode
00:00:19.932398 Drag and drop service loaded
00:00:19.932398 Drag and drop mode: Off
00:00:19.940824 Audio: Detected default audio driver type is 'HostAudioMas'
00:00:19.940824 Error (CPU): 0xC0000005 (STATUS_ACCESS_VIOLATION) xIDP:6ae32489-6ee7-4e23-baed-b257b2a81be8) aComponent={ConsoleMap} aText:[The VBoxGuestPrepSvc service call failed with the error VERR_MSCM_SERVICE_NOT_FOUND], preserve=false aResultDetail=-2900
00:00:19.950409 ***** CFPM dump *****
00:00:19.950426 [J] (level 0)
00:00:19.950438 EnableL0Cap <integer> = 0x0000000000000004 (100)
00:00:19.950437 EnableL0Ext <integer> = 0x0000000000000001 (1)
00:00:19.950442 HWL0Ext <integer> = 0x0000000000000001 (1)
00:00:19.950445 MaxL0ExtSize <integer> = 0x0000000000000008 (8, 8 B)
00:00:19.950446 Name <string> = "Windows 10 (OS)" (cb=16)
00:00:19.950454 MaxCpu <integer> = 0x0000000000000001 (1)
00:00:19.950457 PageL0ExtL0Cap <integer> = 0x0000000000000000 (0)
00:00:19.950459 MaxL0ExtSize <integer> = 0x0000000000000008 (8, 8 B)
00:00:19.950464 MaxL0ExtSize <integer> = 0x0000000000000008 (8, 8 B)
00:00:19.950468 TimerUnit <integer> = 0x0000000000000001 (1)
00:00:19.950471 UUID <bytes> = "fb b6 d6 76 b8 ef 43 ac aa d8 08 66 b6 da fd" (cb=16)
00:00:19.950481 GUI: Cannot notify guest about VM window out-of-focus event
00:00:19.950483
00:00:19.950496 [CPU] (level 1)
00:00:19.950498 EnableL0Ext <integer> = 0x0000000000000001 (1)
00:00:19.950499 GuestCPUName <string> = "host" (cb=5)
00:00:19.950506 HostCPUName <integer> = 0x0000000000000000 (0)
00:00:19.950508 PortableCpuIdLevel <integer> = 0x0000000000000000 (0)
```

Md test: This command creates a directory named test in the current directory.

```
Command Prompt
C:\Users\Addaw\VirtualBox VMs>cd ..
C:\Users\Addaw>md test
C:\Users\Addaw>dir
Volume in drive C is Windows
Volume Serial Number is B29C-B2F3

Directory of C:\Users\Addaw

02/11/2026 12:05 PM <DIR> .
03/17/2025 01:43 AM <DIR> ..
10/27/2025 03:27 PM <DIR> .anaconda
09/25/2024 02:17 PM <DIR> 36 .bash_history
12/12/2025 06:28 PM <DIR> .cache
12/13/2025 06:13 AM <DIR> .conda
10/27/2025 03:18 PM <DIR> 25 .condarc
10/27/2025 03:17 PM <DIR> .continuum
10/04/2024 01:20 PM <DIR> eclipse
08/19/2024 02:06 PM <DIR> 25u .gitconfig
12/12/2025 11:00 PM <DIR> .ipynb_checkpoints
11/16/2025 03:27 PM <DIR> .ipython
11/21/2025 10:40 AM <DIR> .jupyter
11/21/2025 11:17 AM <DIR> .kaggle
12/12/2025 06:28 PM <DIR> .matplotlib
08/20/2025 07:00 PM <DIR> .p2
05/07/2025 09:14 PM <DIR> 176 .packettracer
02/11/2026 11:39 AM <DIR> .VirtualBox
09/25/2024 02:09 PM <DIR> .vscode
10/27/2025 03:19 PM <DIR> .anaconda3
11/03/2025 01:47 PM <DIR> .anaconda_projects
03/22/2024 09:30 PM <DIR> BrainHallaReplays
05/08/2025 06:48 PM <DIR> Cisco Packet Tracer 8.2.2
03/17/2025 01:49 AM <DIR> .Contacts
08/24/2025 08:49 PM <DIR> .Documents
02/09/2026 10:40 PM <DIR> .Downloads
09/18/2024 03:10 PM <DIR> eclipse
08/26/2025 07:01 PM <DIR> eclipse-workspace
03/17/2025 01:49 AM <DIR> .Favorites
02/22/2024 04:33 PM <DIR> .FirstProject
09/11/2024 01:52 PM <DIR> .github
09/18/2024 01:55 PM <DIR> .hazalldr22.github.io
03/17/2025 01:49 AM <DIR> .Links
12/16/2025 12:32 PM <DIR> .Music
08/21/2025 05:26 PM <DIR> .obs-studio
02/10/2026 10:58 AM <DIR> .OneDrive
03/17/2025 01:49 AM <DIR> .Saved Games
03/17/2025 01:49 AM <DIR> .Searches
02/11/2026 12:05 PM <DIR> .test
10/27/2025 03:27 PM <DIR> 72 Untitled.ipynb
10/27/2025 03:27 PM <DIR> 72 Untitled.ipynb
11/21/2025 10:40 AM <DIR> 72 Untitled.ipynb
12/12/2025 11:02 PM <DIR> 628 Untitleds.ipynb
04/21/2025 09:28 PM <DIR> .Videos
02/09/2026 10:47 PM <DIR> .VirtualBox VMs

8 File(s) 1,335 bytes
37 Dir(s) 271,895,986,176 bytes free

C:\Users\Addaw>
```

Cd test: This command changes the working directory to the test directory we just made.

```
C:\Users\Addaw>cd test  
C:\Users\Addaw\test>
```

(Cd ..): This command brings the working directory one backwards in the file path.

```
C:\Users\Addaw\test>cd ..  
C:\Users\Addaw>
```

Rd test: Removes the test directory we just made. I also used the dir command to show that it was removed.

```
Command Prompt
C:\Users\Addaw\test>cd ..
C:\Users\Addaw>rd test
C:\Users\Addaw>dir
Volume in drive C is Windows
Volume Serial Number is B29C-B2F3

Directory of C:\Users\Addaw

02/11/2026  12:15 PM  <DIR>      .
03/17/2025  01:43 AM  <DIR>      ..
10/27/2025  03:27 PM  <DIR>      .anaconda
09/25/2024  02:17 PM          36 .bash_history
12/12/2025  06:28 PM  <DIR>      .cache
12/13/2025  06:13 AM  <DIR>      .conda
10/27/2025  03:18 PM          25 .condarc
10/27/2025  03:17 PM  <DIR>      .continuum
10/04/2024  01:20 PM  <DIR>      .eclipse
09/18/2024  02:06 PM          254 .gitconfig
12/12/2025  11:00 PM  <DIR>      .ipynb_checkpoints
11/10/2025  03:27 PM  <DIR>      .ipython
11/21/2025  10:44 AM  <DIR>      .jupyter
11/21/2025  11:17 AM  <DIR>      .kaggle
12/12/2025  06:28 PM  <DIR>      .matplotlib
04/20/2025  07:00 PM  <DIR>      .p2
05/07/2025  09:14 PM          176 .packetttracer
02/11/2026  11:39 AM  <DIR>      .VirtualBox
09/25/2024  02:09 PM  <DIR>      .vscode
10/27/2025  03:19 PM  <DIR>      anaconda3
11/03/2025  01:47 PM  <DIR>      anaconda_projects
03/22/2024  09:30 PM  <DIR>      BrawlhallaReplays
05/08/2025  06:48 PM  <DIR>      Cisco Packet Tracer 8.2.2
03/17/2025  01:49 AM  <DIR>      Contacts
04/24/2025  08:49 PM  <DIR>      Documents
02/09/2026  10:40 PM  <DIR>      Downloads
09/10/2024  03:10 PM  <DIR>      eclipse
04/20/2025  07:01 PM  <DIR>      eclipse-workspace
03/17/2025  01:49 AM  <DIR>      Favorites
02/22/2024  04:33 PM  <DIR>      FirstProject
09/11/2024  01:52 PM  <DIR>      github
09/18/2024  01:55 PM  <DIR>      hazalldr22.github.io
03/17/2025  01:49 AM  <DIR>      Links
12/16/2025  12:32 PM  <DIR>      Music
04/21/2025  05:26 PM  <DIR>      obs-studio
02/10/2026  10:58 AM  <DIR>      OneDrive
03/17/2025  01:49 AM  <DIR>      Saved Games
03/17/2025  01:49 AM  <DIR>      Searches
10/27/2025  03:27 PM          72 Untitled.ipynb
10/27/2025  03:27 PM          72 Untitled1.ipynb
11/21/2025  10:44 AM          72 Untitled2.ipynb
12/12/2025  11:02 PM          628 Untitled3.ipynb
04/21/2025  09:28 PM  <DIR>      Videos
02/09/2026  10:47 PM  <DIR>      VirtualBox VMs
               8 File(s)          1,335 bytes
              36 Dir(s)  271,894,216,704 bytes free

C:\Users\Addaw>
```

Ipconfig: This command outputs the network adapter settings. It shows important information such as the IPv4 address.

```
Command Prompt
36 Dir(s) 271,894,216,704 bytes free
C:\Users\Addaw>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Ethernet adapter Ethernet 4:

    Connection-specific DNS Suffix . :
    Link-local IPv6 Address . . . . : fe80::3432:5444:9366:b7bd%21
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Unknown adapter Local Area Connection 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Unknown adapter OpenVPN Connect DCO Adapter:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Ethernet adapter Ethernet:

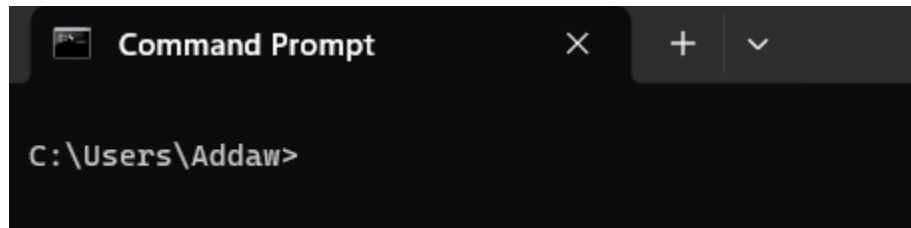
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : sbu.edu
    Link-local IPv6 Address . . . . : fe80::9094:8d3:97b8:afba%22
    IPv4 Address. . . . . : 10.8.211.152
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 10.8.0.1

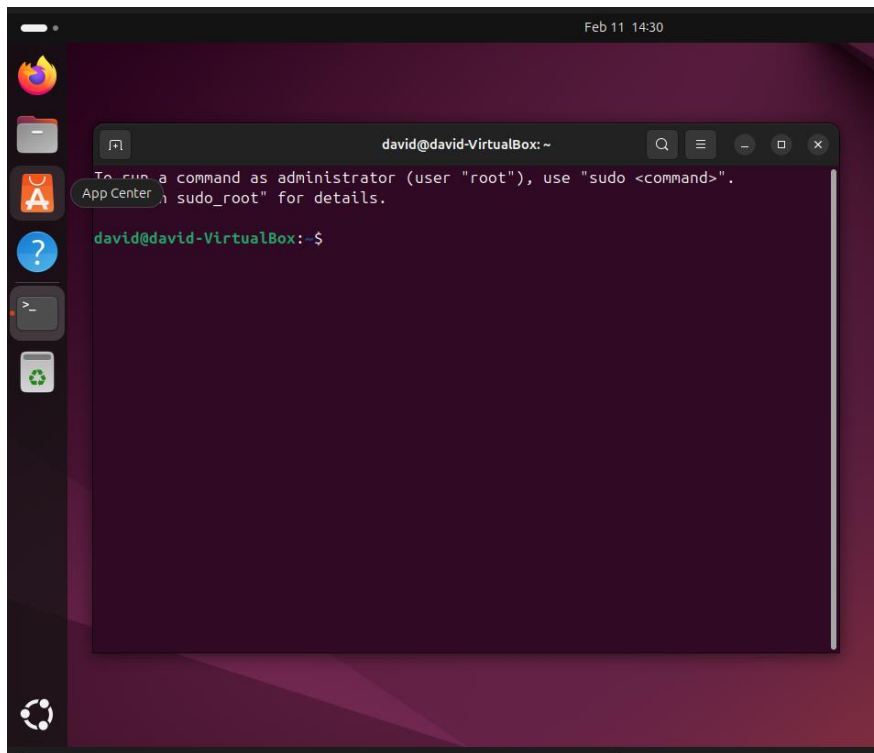
C:\Users\Addaw>
```

Cls: This command clears everything that was output in the current cmd session.

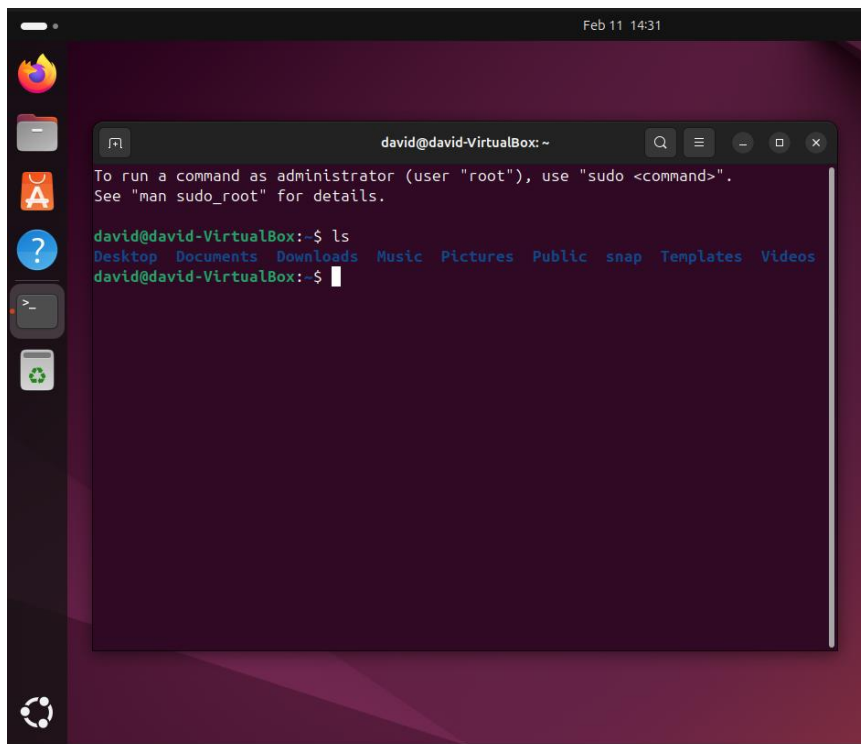


Using VM Linux:

Linux command line:



Ls: Lists the contents of the current directory.



Cat: This stands for concatenate, which when called outputs the contents of a file. In my case, I created a test file in my test directory so that the cat command would run. I have nothing in the file, so nothing is output.

```
david@david-VirtualBox:~$ cd test
david@david-VirtualBox:~/test$ touch testFile.txt
david@david-VirtualBox:~/test$ ls
testFile.txt
david@david-VirtualBox:~/test$ cat testFile.txt
david@david-VirtualBox:~/test$
```

Mkdir test: Creates the test directory.

```
david@david-VirtualBox:~$ mkdir test
david@david-VirtualBox:~$ ls
Desktop    Downloads  Pictures   snap       test
Documents  Music      Public     Templates  Videos
```

Cd test: Changes the working directory to test.

```
david@david-VirtualBox:~$ cd test
david@david-VirtualBox:~/test$
```

(Cd ..): This command brings the working directory one backwards in the file path. (Same as cmd)

```
david@david-VirtualBox:~/test$ cd ..
david@david-VirtualBox:~$
```

rm test and rmdir test: In my case I made a file in the test directory, so I had to use “rm testFile” before I could remove the directory. Then, I removed the directory using “rmdir test.”

```
david@david-VirtualBox:~$ cd test
david@david-VirtualBox:~/test$ rm testFile.txt
david@david-VirtualBox:~/test$ ls
david@david-VirtualBox:~/test$ cd ..
david@david-VirtualBox:~$ rm test
rm: cannot remove 'test': Is a directory
david@david-VirtualBox:~$ rmdir test
david@david-VirtualBox:~$ ls
Desktop Documents Downloads Music Pictures Public snap Templates Videos
david@david-VirtualBox:~$
```

Ifconfig (ip address): On the updated Ubuntu VMs, ifconfig isn't installed by default so I used "ip address" which works the same. This command similarly outputs important network adapter information.

```
david@david-VirtualBox:~$ ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:48:b7:0c brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 80259sec preferred_lft 80259sec
    inet6 fd17:625c:f037:2:4a31:f20e:cc80:d12a/64 scope global temporary dynamic
        valid_lft 86399sec preferred_lft 14399sec
    inet6 fd17:625c:f037:2:a00:27ff:fe48:b70c/64 scope global dynamic mngtmpaddr
        valid_lft 86399sec preferred_lft 14399sec
    inet6 fe80::a00:27ff:fe48:b70c/64 scope link
        valid_lft forever preferred_lft forever
david@david-VirtualBox:~$
```

Clear: Clears all previous prompts and information.

